COLLEGE OF NATURAL SCIENCES AND MATHEMATICS

Second Semester, Academic Year 2025-2026

University Vision	**:**	A transformative University driving excellence, research, innovation, and sustainable peace and development
University Mission	**:**	To produce competent human resources committed to the development of the Southern Philippines and to help uplift the living conditions of Muslims, Indigenous Peoples and other underserved communities
University Core Values	**:**	Integrity, Respect, Inclusivity, Service, Excellence
College Goals	**:**	1. To provide an excellent educational and training platform for undergraduate and professional students in Chemistry, Physics, Biology, Mathematics, Information Technology, and Sustainable Development Studies equipping them with the knowledge, skills, and values necessary for careers in research and teaching, the private industry, and the government sector, as well as for the pursuit of advanced studies. 2. To produce highly skilled, competent, and competitive graduates by enhancing their capacities in critical thinking and analysis, problem solving, technical writing, oral communication, interpersonal relations, and strategic foresight and futures thinking. 3. To enable and capacitate students to adapt to and perform productively in diverse and challenging work environments by instilling in them the values of hard work, perseverance, excellence, integrity, independence, teamwork, discipline, responsibility, and accountability, as well as a strong sense of prudent, fact-based, and objective decision-making anchored on the principles of sustainable development. 4. To train students in conducting basic and applied research, as well as in organizing extension work with community beneficiaries to empower the students and build a strong sense of community and kinship, and in the process become catalysts of change in their respective communities, and champions of sustainability.

Program Educational Objectives	:	1. Produce graduates who comply with the current qualification requirements of professional chemists for local and overseas employment and entrepreneurship. 2. Prepare students for higher studies in chemistry and in other fields.

Course Title	:	Inorganic Chemistry 1
Course Number	:	CHM151
Time Allotment	:	54 hours (3 hours Lecture/week)
Credit Units	:	3 units
Prerequisites	:	CHM100, CHM100.2
Course Description	:	This course is devoted to the study of the principles and trends in the chemistry of the elements. Topics also include electrochemistry, reduction-oxidation reactions, nuclear chemistry and descriptive chemistry of metals and nonmetals.
Values Integration	:	Excellence: Students strive for mastery in understanding periodic trends and chemical principles, ensuring high standards in their scientific analysis. Integrity: Students uphold honesty and accuracy in reporting scientific data and observations regarding chemical properties.
SDG Integration	:	SDG 7: Affordable and Clean Energy: By studying electrochemistry and reduction-oxidation reactions, students gain foundational knowledge essential for the development of energy storage systems and batteries. SDG 12: Responsible Consumption and Production: The study of metal chemistry and extraction processes fosters an understanding of material lifecycles and the importance of sustainable resource management.

Program Outcomes	PEO1	PEO2
PO1: Synthesize knowledge from the core areas of chemistry to formulate scientifically viable solutions to multifaceted chemical problems.	✓	✓
PO2: Solve chemical problems of varying complexity using core and interdisciplinary scientific approaches to advance research, peace, and sustainability.	✓	✓
PO3: Synthesize emerging knowledge and skills with foundational principles to prepare for graduate studies or entry-level professional roles.	✓	✓
PO4: Evaluate the ethical, environmental, and global implications of chemical practices and innovations to promote sustainability and responsible citizenship.	✓	✓
PO5: Produce professional scientific communications in English and Filipino that meet disciplinary standards.	✓	
PO6: Design regionally relevant experimental procedures using modern instrumentation and digital tools to acquire accurate chemical data.	✓	✓
PO7: Develop innovative and feasible chemistry-based solutions or products by considering market/community needs and the sustainable use of regional resources.	✓	
PO8: Create sustainable chemical solutions that integrate ethical standards and cultural preservation for the benefit of marginalized communities.	✓	
PO9: Lead multidisciplinary teams to implement chemistry-related initiatives that foster innovation and excellence.	✓	

Code: Put a check mark (✓) in the appropriate box if this PO addresses which particular PEO.

Course Outcomes	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**
CO1: Analyze the fundamental principles of atomic structure, quantum mechanics, chemical bonding, and states of matter to predict molecular geometries and physical properties of substances.	L	L	L			L			
CO2: Apply stoichiometric principles and thermodynamic concepts to solve quantitative problems involving reduction-oxidation systems, electrochemical cells, and nuclear transformations.	P	P	P	L		P			
CO3: Evaluate periodic trends and the descriptive chemistry of elements to explain chemical behavior, reactivity, and their implications in industrial and environmental contexts.	D	D	P	P		P			

Code: L: Upon attainment of this CO, students will have learned the theories, principles and concepts of which PO; **P**: Upon attainment of this CO, students will have practiced which PO with maximum supervision; **D**: Upon attainment of this CO, students will have demonstrated which PO with minimum or no supervision.

Course Outcomes	**SDG**																
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**
CO1: Analyze the fundamental principles of atomic structure, quantum mechanics, chemical bonding, and states of matter to predict molecular geometries and physical properties of substances.																	
CO2: Apply stoichiometric principles and thermodynamic concepts to solve quantitative problems involving reduction-oxidation systems, electrochemical cells, and nuclear transformations.							IC										

CO3: Evaluate periodic trends and the descriptive chemistry of elements to explain chemical behavior, reactivity, and their implications in industrial and environmental contexts.									IC			IC					

SDGs: 1: No Poverty; **2:** Zero Hunger; **3:** Good Health and Well-being; **4:** Quality Education; **5:** Gender Equality; **6:** Clean Water and Sanitation; **7:** Affordable and Clean Energy; **8:** Decent Work and Economic Growth; **9:** Industry, Innovation, and Infrastructure; **10:** Reduced Inequalities; **11**: Sustainable Cities and Communities; **12:** Responsible Consumption and Production; **13:** Climate Action; **14:** Life Below Water; **15:** Life on Land; **16:** Peace, Justice, and Strong Institutions; **17:** Partnerships for the Goals.

Code: DC: The course directly contributes to the SDG through its activities and or outputs designed to explicitly and intentionally achieve the goal; **IC**: The course indirectly contributes to the SDG through its activities and or outputs that provide minimal or secondary support to the goal.

INSTRUCTIONAL PLAN

Learning Outcome	CO1	CO2	CO3	Content	Flexible Teaching-Learning Strategies			Assessment	Timeframe
					Onsite	Online	Offline		
COURSE ORIENTATION									
1. Analyze the evolution of atomic theory and apply quantum mechanical principles to determine electronic configurations and predict periodic trends.	L		L	**1. Atomic Structure** a. Evolution of atomic theory, discovery of electron and nucleus b. Failure of classical mechanics, and the need for quantum mechanics c. Wave-particle duality of light and the photoelectric effect d. Wave-particle duality of matter and the Schrödinger equation e. Hydrogen atom energy levels	Interactive lecture on quantum numbers and orbital shapes.	Virtual simulation of the photoelectric effect and atomic orbitals.	Problem set on determining electron configurations and identifying periodic trends.	**Traditional:** Quiz on quantum numbers and electronic configurations. **Authentic:** Group analysis of periodic trend anomalies.	Week: 1-5 Hours: 15

				f. Hydrogen atom wavefunctions (orbitals) and quantum numbers g. s-,p-, and d-orbitals h. Multielectron atoms and electron configuration i. Periodic trends					
FIRST EXAMINATION: Date									
2. Compare ionic and covalent bonding models and apply VSEPR and Molecular Orbital theories to predict molecular geometry and bonding properties.	**P**			**2. Molecular Structure** a. Introduction to chemical bonds, type of bonds b. Covalent bond c. Octet rule and Lewis structure d. Exception to Lewis structure rules (less than/more than octet, odd number of electrons) e. Polar covalent bond f. Ionic bond g. VSEPR theory h. Molecular orbital theory i. MOT of simple diatomic molecule j. MOT of triatomic molecules k. Valence bond theory and hybridization l. Metallic bond	Molecular model building activity to visualize VSEPR geometries.	Video demonstration of hybridization and MO diagrams.	Worksheet on drawing Lewis structures and assigning formal charges.	**Traditional:** Problem solving on VSEPR theory and hybridization. **Authentic:** Construction of molecular models for selected compounds.	Week: 6-9 Hours: 12
SECOND EXAMINATION: Date									

3. Differentiate the properties of gases, liquids, and solids based on kinetic molecular theory and intermolecular forces.	**D**			**3. States of Matter** a. Gas laws b. Kinetic molecular theory c. Solids and liquids and intermolecular forces d. Phase change and phase diagram e. Properties of liquid state f. Structure, properties and bonding of solids	Problem-solving session on gas laws and phase diagrams.	Interactive module on intermolecular forces and phase changes.	Reading assignment on the industrial applications of phase diagrams.	**Traditional:** Calculation test on gas laws and kinetic energy. **Authentic:** Case study analysis of a phase diagram for a specific substance.	Week: 10-13 Hours: 12
THIRD EXAMINATION: Date									
4. Analyze oxidation-reduction reactions and electrochemical cells to determine standard potentials and thermodynamic feasibility.		**P**	**D**	**4. Oxidation and Reduction** a. Introduction to oxidation and reduction reactions, oxidation states b. Balancing oxidation-reduction reactions c. Standard electrode potential and thermodynamics of redox reactions d. Disproportionation reactions e. Potential diagram (Latimer, Frost, Pourbaix diagram) f. Application of redox reactions to the extraction of elements	Problem-solving workshop on balancing redox equations and calculating cell potentials.	Video demonstration of electrochemical cells and potential diagrams.	OWorksheet on constructing Latimer and Frost diagrams.	**Traditional:** Quiz on balancing redox reactions and Nernst equation calculations. **Authentic:** Investigation of corrosion prevention methods.	Week: 14-16 Hours: 9

5. Evaluate nuclear stability and kinetics of radioactive decay to assess applications and hazards of radioisotopes	**L**	**P**		**5. Nuclear Chemistry** a. Radioactive decay and nuclear stability b. Kinetics of radioactive decay c. Nuclear transmutation d. Applications of radioisotopes e. Radioactivity hazards	Interactive lecture on nuclear stability belts and decay modes.n	Virtual simulation of radioactive decay kinetics.	Case study reading on medical applications of radioisotopes.	**Traditional:** Problem set on half-life calculations and nuclear equations. **Authentic:** Presentation on the environmental impact of nuclear waste.	Week: 17-18 Hours: 6
COURSE INTEGRATION: Date									
FINAL EXAMINATION: Date									

Code: **L** = Upon attainment of this LO, students will have learned the theories, principles and concepts of which CO; **P** = Upon attainment of this LO, students will have practiced which CO with maximum supervision; **D** = Upon attainment of this LO, students will have demonstrated which CO with minimum or no supervision.

COURSE REQUIREMENTS:

Attendance: Regular attendance in lecture sessions is required.
Examinations: Completion of three (3) Departmental/Major Examinations and one (1) Final Examination.
Quizzes: Passing short quizzes administered throughout the semester.
Problem Sets: Submission of assigned problem sets and worksheets to demonstrate understanding of quantitative concepts.

GRADING COMPONENTS:

Component	Weight (%)
Problem Sets	10
Quizzes	20
Exams	70
Total	100

GRADING SYSTEM:

Rating (%)	Grade	Level of Competence
96 – 100	1.00	Excellent
91 – 95	1.25	Excellent
86 – 90	1.50	Very Good
81 – 85	1.75	Very Good
76 – 80	2.00	Good
71 – 75	2.25	Good
66 – 70	2.50	Satisfactory
61 – 65	2.75	Satisfactory
50 – 60	3.00	Passing
Below 50.00	5.00	Failure
	INC	Incomplete
	DRP	Dropped
	W	Withdrawn
	INP	In Progress

REFERENCES:

1. Atkins, P., & Jones, L. (2007). *Chemical principles: The quest for insight* (4th ed.). W.H. Freeman and Company.
2. Brown, T. L., LeMay, H. E., Bursten, B. E., Murphy, C. J., Woodward, P. M., & Stoltzfus, M. W. (2015). *Chemistry: The central science* (13th ed.). Pearson Education, Inc.
3. Petrucci, R. H., Harwood, W. S., & Herring, F. G. (2007). *General chemistry: Principles and modern applications* (9th ed.). Prentice-Hall International, Inc.
4. Silberberg, M. S., & Amateis, P. G. (2015). *Chemistry: The molecular nature of matter and change* (7th ed.). McGraw Hill Education.

Additional Guidelines and Recommendations

Mastery of Periodic Trends: Inorganic chemistry relies heavily on the patterns within the Periodic Table. Focus on understanding the underlying physical reasons (e.g., shielding effect, effective nuclear charge) for these trends rather than just memorizing them, as this will allow you to predict chemical behavior in unknown scenarios.

Spatial Visualization: Many topics, particularly Molecular Structure (VSEPR, Molecular Orbital Theory) and Solid State Chemistry (Unit Cells), require strong 3D visualization skills. Using molecular modeling kits or visualization software is highly recommended to grasp these spatial relationships effectively.

Mathematical Proficiency: Quantitative analysis is crucial in this course, especially for Electrochemistry (Nernst equation), Nuclear Chemistry (kinetics of decay), and Gas Laws. Practice solving complex problems systematically, paying close attention to units and significant figures.

Connecting Theory to Reality: Strive to connect abstract concepts like Quantum Mechanics and Bonding Theories to observable physical properties of matter (e.g., melting points, conductivity, magnetic properties). This conceptual linking is vital for deep understanding.

Systematic Approach to Redox: Balancing oxidation-reduction reactions, especially in acidic or basic media, requires a step-by-step systematic approach. Regular practice of the ion-electron method is essential to avoid common pitfalls during exams.

COURSE POLICIES, REGULATIONS, AND SPECIAL NEEDS:

A. Schedule of Class

Our class will be flexible and may be conducted either face-to-face or remotely, with a total of 54 instructional hours. At the beginning of the term, a course orientation will be conducted to clarify course objectives, requirements, learning outcomes, assessment tasks, and class expectations. Throughout the semester, scheduled course integration, in addition to the regular consultation schedule, will serve as dedicated period for discussing your academic standing, course clarifications, performance feedback, and other academic concerns. If a regular class meeting needs to be canceled, I will provide alternative instructions and supplementary learning activities to ensure continuity and help you stay on pace. Remedial classes may also be arranged, when necessary, to support students who need additional guidance. The examination schedule will follow the official university calendar, although adjustments may be considered in coordination with your other academic commitments. Any changes to the class schedule will be communicated in advance through our agreed-upon communication platform, such as Messenger.

B. Student Attendance

It is your solemn duty to attend face-to-face or remote classes according to the stipulated class schedule. Relative to such a duty, punctuality is a highly appreciated value. Thus, being tardy for a fairly enough time of eighteen (18) minutes or 20% of the one and half hour allotted for each meeting will be considered an absent attendance. Any additional attendance requirements will be considered on a case to case basis since the department endeavors to be inclusive to all, including those with disabilities and other valid circumstances as we may determine. While the department is flexible and will exhaust all means to show consideration and understanding of your individual case, you are responsible for compliance of missed works. Thus, when I have requested, you should provide a waiver or excuse letter for possible absence. In the event of an excused absence, you should produce proper documentation, such as a health certificate or a parent note, and should make arrangements with me to make up any missed work. Attendance is highly regarded in classes since they affect your final grades. If you missed nearly seven cumulative meetings, I will inform you that you will be dropped in the class on the seventh.

C. Religious and Cultural Observance

Religious and cultural observances are an essential reflection of diversity. I am committed to providing equitable educational opportunities to all. At the beginning of the semester, you should review the class requirements to identify possible conflicts with classes and other required activities. Those who have religious or cultural observances that coincide with the classes and other required activities should inform me by writing within the first two weeks of the first meeting to allow time for us to discuss and make equitable adjustments. I strongly encourage you to honor your cultural and religious holidays. However, if I do not hear from you, I will assume that you plan to attend the class.

D. Diversity in Points of View

All diverse points of view are hoped to be well served in this class, and such diverse points of view in this class are seen as an advantage, resource, and strength. I am committed to providing an atmosphere of learning that is representative of a variety of perspectives. The ideas in the class will inevitably suggest some particular points of view which do not have to be yours. I encourage you to disagree with these points of view as well as those of your classmates. Please feel free to express yourself. A significant part of state university education is learning about the complexity of various perspectives. Thus, we must respect one another, but we do not necessarily have to agree. A richer discussion will occur when a variety of points of view are presented on the table for discussion.

E. Discussion Guidelines

Guidelines for inclusive conversation are critical for creating a safe and inclusive learning environment. To achieve this, it is critical to adhere to specific guiding principles that promote diversity. First and foremost, always actively listen to people and encourage each other to offer their perspectives. This includes not interrupting or talking over others, as well as not assuming their thoughts or experiences. It is also critical to employ inclusive language that does not exclude someone on the basis of their gender, color, religion, or other characteristics. Avoid making offensive jokes or using slurs or abusive words. If you're not sure what terminology to use, ask for clarification or research the topic beforehand. Another key rule is to avoid criticizing someone's character or views during a discussion. Instead, stay focused on the topic at hand and use respectful language even when you disagree with someone. It's critical to acknowledge that everyone comes from a unique set of circumstances and experiences that affect their viewpoint. Try to respect these differences and recognize that there is no one “correct” method of thinking. Finally, be willing to learn from others and admit when you've made a mistake. We are all always learning and growing, therefore making errors is acceptable as long as we are prepared to listen and learn from others. By following these guidelines, we can create an inclusive and welcoming environment where everyone feels heard and valued.

F. Accommodation and Accessibility

The department endeavors to be inclusive and to give equal opportunities to all with disabilities. If you have some form of disability, please inform me at your earliest convenience so that I can accommodate, adapt, personalize, or modify my instruction to facilitate your learning needs. Accommodations are provided at no cost if you are eligible. We will try our best to give you reasonable accommodations so that you can participate and progress in the class. To request for reasonable accommodation, fill out the necessary details in the letter of accommodation. This letter can be requested at the office of the department chairperson. The department will treat your information with utmost confidentiality. Please note that while the accommodation policies hope to be inclusive to all, these do not intend to lower academic standards or compromise the integrity of the academic program. The academic conduct and technical standards of the department will be upheld at all times.

G. Universal or Inclusive Design for Learning

You are highly encouraged to actively participate in classroom activities, discussions, and assignments to demonstrate your understanding of the lesson and contribute to learning. You should notify me as soon as possible if they are unable to attend or participate in class for whatever reason, and you should collaborate with me to make up for missing work or assignments. I will try my best to make reasonable accommodations and give assistance for equitable access to educational opportunities. Moreover, to promote universal or inclusive learning, I will use all available resources to ensure that accommodations, assistance, adaptations, or modifications, such as accessible learning materials, are accessible to you, including those with disabilities. I will also provide flexible training that includes numerous forms of instruction such as visual aids, hands-on exercises, and interactive conversations. Additionally, I will use positive reinforcement to promote and celebrate performance and development, such as recognition for academic achievement or engagement in related activities. Finally, I will develop a culture of respect for diversity and inclusion in which all, regardless of color, ethnicity, gender identity, sexual orientation, religion, or other personal traits, feel appreciated and supported.

H. Student Mental Health and Well-being

Mental and emotional disturbance, including stress, anxiety, mood changes, diminished self-esteem, or problems with eating and or sleeping can affect your academic performance. The source of symptoms might be strongly linked to your academic endeavors; if so, please feel free to approach me. You may also visit the proper office that provides mental health and counseling services to help you cope with difficult emotions and life stressors that may threaten your well-being. The services are free and completely confidential. If you find yourself feeling distressed and in need of more immediate support, I encourage you to seek help. You may contact our University Office of Guidance Counselor thru the following details: 09088105773 / guidance@msugensan.edu.ph.

I. Materials Ownership and Confidentiality

All instructional materials e.g., slide presentations, digital worksheets, etc. given to you are my sole properties, thus the right to ownership cannot be transferred to you or any other person. Such materials should be treated with confidentiality and should never be reproduced electronically such as forwarding it to others or uploading copies to online databases without oral and written consent of me. You have the responsibility to secure the confidentiality of these materials that should only be utilized for your individual scholarly consumption.

J. Student Discipline in All Environments

A face-to-face and online or remote environment is under my authority. I would like you to take the opportunity to demonstrate civil behavior, proper etiquette, and appropriate respect to everyone in our physical classroom and chosen social media communication platform. I encourage you to express your freedom of expression or your opinions and comments relative to a particular discussion or task, but please do so with respect and sensitivity to the ideas and feelings of everyone. Derogatory comments, ranting, and vulgar language are not acceptable. Virtually, you are encouraged to use your real full names in your online accounts so that I can easily identify you. Upon information, you are expected to download learning materials if needed and once available. When virtually submitting requirements or consulting me, take it as opportunity to observe and practice general standards of written communication. In our physical classroom, you should ensure the cleanliness and orderliness during and before exiting the class. In terms of submissions, if you commit any form of cheating, you will be given a warning and shall be treated according to university policies. Plagiarism or copying words, ideas, or arguments from another person without citation in any of the class requirements submitted will not be tolerated. It is your responsibility to monitor your current grade standing in the class and to inform me regarding your assessment concerns and questions.

K. Artificial Intelligence Use

This class maintains a policy that you may use artificial intelligence (AI) to assist you in the writing process but that all artificially-generated text needs to be explicitly labeled. In handing in your assignment, you agree to disclose the extent to which you used ChatGPT or other AI writing tools in your assignment. All text by AI must be quoted with the source of the model in parenthesis (ChatGPT). At the end of your paper please include the following example statement. Failure to adequately disclose your AI use will result in a zero for the assignment. Example statement: “This paper used AI for the following components of writing process: (choose from the following: brainstorming, editing, sentence generation).”

L. Observance of Green Practices

The term “green practices” pertains to actions that decrease the environmental impact of wastes. Relative to such waste concern, this class promotes a zero-waste class. As its little way to support and contribute to the cause of a green environment, you are encouraged if possible to use digital copies of the instructional materials, instructed to email most of the requirements, take traditional assessments via online, and use recyclable materials e.g., stainless container for water and food, eco bag for safekeeping of stuff, etc.

M. Sanction Clauses

The sanctions in this class shall apply to anyone who violates our stipulated policies or regulations. Upon consultation with relevant authorities and contemplation of all involved factors, I may decide to take an appropriate course of action especially for serious violations related but not restricted to student attendance, student discipline, virtual etiquettes, academic integrity, and materials ownership.

FACULTY INFORMATION:

Name: Ramon M. Eduque, Jr.
Email Address: ramon.eduque@msugensan.edu.ph
Mobile Number: Kindly contact me through email.
Office: Department of Chemistry, RSRC, CNSM Complex

CONSULTATION SCHEDULE:
TFr, 1:00 pm to 4:00 pm, Faculty Office
W, 10:00 am to 3:00 pm, Faculty Office

Prepared by:	Evaluated by	Recommended by:	Endorsed by:
RAMON M. EDUQUE, JR. Course Adviser	**THUCYDIDES L. SALUNGA, PHD** DCC Representative	**RONALDO T. BIGSANG, MS** Department Chair	**JAY D. BUSCANO, PHD** Dean

Checked by:	Approved by:
MICHAEL B. CAHAPAY, PHD CCSID Director	**MISHELL D. LAWAS, D. Engr.** VCAA

OBE SYLLABUS RECEIPT FORM

By signing this form, you confirm that you have read, understood, agreed, and received a copy of the syllabus for the course detailed below:

Course Code: **CHM151**
Course Title: **Inorganic Chemistry 1**
Course Description: **This course is devoted to the study of the principles and trends in the chemistry of the elements. Topics also include electrochemistry, reduction-oxidation reactions, nuclear chemistry and descriptive chemistry of metals and nonmetals.**
Unit Credits: **3 units**
Instructor: **Ramon M. Eduque, Jr.**
Semester and Academic Year: **Second Semester, AY 2025-2026**

☑ I have been oriented on the vision, mission, and core values of Mindanao State University – General Santos; goals of the College of Natural Sciences and Mathematics; and objectives of Bachelor of Science in Chemistry.

☑ I have been informed of the intended learning outcomes of the course, instructional resources, assessment tasks, course requirements, grading system, together with the course policies and regulations that I am expected to observe and comply with throughout the duration of the semester.

☑ I have been given a complete copy of the official course syllabus, either in electronic or printed format, which will serve as my primary reference and guide throughout the duration of the course.

	Student Name	Signature	Date Received
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